# Fewer Giants, More Firms?
# Artificial Intelligence, Entry Costs, and the Firm Size Distribution*

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**Abstract**

Can firm formation induced by artificial intelligence (AI) offset the labor that large incumbents shed as they adopt it? We study a partial-equilibrium firm-dynamics model in which AI is both a labor-saving input at adopting firms and a cut in the cost of entry. Wages are fixed, so every outcome is a change in labor demand, not in equilibrium employment, and the model is not calibrated. Whether an adopter hires or sheds workers is the Hicks–Marshall comparison of the labor–AI substitution elasticity with the firm's scale elasticity; with span-of-control returns $\alpha = 0.64$ the threshold is 2.78, but it rises steeply with $\alpha$. Holding the cost saving from AI fixed as the elasticity varies, the net labor-demand effect declines monotonically in the elasticity: offset fails when substitution is high and AI-native entrants are no more productive than earlier entrants, and cheaper entry matters little. In Census business applications, two highly AI-exposed sectors pulled away from other sectors from March 2025, by 0.29 log points in 2026, mostly among applicants unlikely to hire; the estimate is sensitive to which sectors count as exposed.

**JEL codes:** D22, E24, J23, L11, L26, O33.
**Keywords:** artificial intelligence, cloud computing, firm entry, firm size distribution, business dynamism, labor displacement.

*The views expressed are the author's own and do not represent those of any employer. Any errors are the author's own. The data, code, and replication files are available at https://github.com/gspivey/ai-employment-offset. Large-language-model tools were used to assist with coding and editing; all economic content, modelling choices, and interpretations are the author's own.

# 1 Introduction

Firms are adopting artificial intelligence (AI) rapidly, and early experimental and field evidence documents sizable productivity gains in specific tasks (Noy and Zhang, 2023; Brynjolfsson et al., 2025b). If these gains are labor-saving at the firm level, the first-order prediction is that incumbents will produce the same output with fewer workers (Acemoglu and Restrepo, 2019). The concern is sharpest for large firms, which have the data, capital, and organizational capacity to adopt early: in the European Union in 2025, 55% of enterprises with 250 or more employees used AI, against 17% of those with 10 to 49 employees (Eurostat, 2026b). This paper is about the second-order response. A technology that makes existing firms more productive may also make new firms cheaper to create. We ask under what conditions, when AI reduces labor demand at large incumbents, it also lowers the barriers to firm formation enough that labor demand at smaller firms and entrants offsets the loss.

The answer matters for how economists and policymakers read the labor-market consequences of AI. Most existing assessments measure the exposure of occupations and tasks (Eloundou et al., 2024; Acemoglu, 2025), implicitly holding the set of firms fixed. Yet net job creation in the United States is driven disproportionately by young firms rather than by small firms as such (Haltiwanger et al., 2013), and the long decline in business dynamism has been linked to slower reallocation and weaker productivity growth (Decker et al., 2014, 2020). If AI moves the entry margin, changes in the number and size of firms could rival the within-firm displacement that task-based accounts emphasize.

Answering the question is hard for three reasons. First, AI is still diffusing, so the post-shock firm size distribution is not yet observed and any direct test is necessarily partial. Second, the two channels operate at different speeds: incumbents can cut headcount within a budget cycle, whereas the entry, survival, and scale-up of new firms unfold over years. Third, the natural historical comparison is not clean: the cloud era coincided with a secular decline in the aggregate US startup rate (Decker et al., 2016; Akcigit and Ates, 2021), much of it plausibly demographic (Karahan et al., 2024; Hopenhayn et al., 2022), so any entry effect of cloud computing would have to be identified within exposed sectors and against a falling baseline.

We study the question in a partial-equilibrium model of firm dynamics with entry and exit (Section 2), in which the AI shock lowers adopters' unit costs through a CES bundle of labor and AI services, lowers the entry cost, and may make entrants more productive (Section 4). Two features of the exercise frame every result. Wages are fixed and product demand is not modeled, so outcomes are changes in *labor demand* at a constant wage, not equilibrium employment. And the model is not calibrated: its parameters are illustrative defaults, and Section 5.1 reports how far its pre-AI moments are from US data.

We obtain three results. First, at a single firm the sign of the headcount response is the textbook Hicks–Marshall comparison of substitution and scale effects (Hamermesh, 1993): AI raises an adopter's labor demand if and only if the labor–AI elasticity of substitution $\sigma_{\text{ai}}$ is below the scale elasticity (Proposition 1). Under span-of-control returns $\alpha$ the scale elasticity is $1/(1-\alpha)$, which is 2.78 at $\alpha = 0.64$ but 6.67 at $\alpha = 0.85$, a value closer to estimates of returns to scale at the establishment level (Atkeson and Kehoe, 2005; Restuccia and Rogerson, 2008); the threshold, and hence the regime, is therefore only as well pinned down as $\alpha$ (Table 2). Second, in the original specification $\sigma_{\text{ai}}$ was confounded with the size of the shock: with the price of AI held fixed, a low elasticity is also a very large cost reduction. Once the cost saving at a reference point is held fixed as $\sigma_{\text{ai}}$ varies (Klump and de La Grandville, 2000; León-Ledesma et al., 2010), the net labor-demand effect declines monotonically in $\sigma_{\text{ai}}$, and offset fails at high substitution unless AI-native entrants are substantially more productive (Figure 1). The flat "plateau" at high $\sigma_{\text{ai}}$ in earlier drafts was an

artifact of the fixed AI price. Across specifications, cutting the entry cost moves aggregate labor demand little; it changes where employment sits more than how much of it there is. Third, in Census business applications, Information and Professional services pulled away from other sectors from March 2025, with no pre-trend, mostly among applications with a low propensity to hire (Section 6). That pattern is consistent with more but smaller entrants, and also with displaced workers registering as contractors; it is sensitive to which sectors are classed as exposed, and we do not give it a causal reading.

The cloud-computing era (2006–2020) serves as motivation rather than as a source of discipline (Section 3). Cheaper computing coincided with rising employment concentration and a falling startup rate. The model's best-fitting configuration for those facts *raises* the employer entry cost by 20%, which reverses the sign of the entry-cost cut the cloud analogy was meant to represent, so we do not use the replication to inform the AI priors. In the discussion (Section 7) we note that leading AI application firms are far more employment-light than cloud-era software firms, consistent with systematic evidence that AI-native startups are smaller (Kim and Koning, 2026).

### 1.1 Related literature

A large literature studies displacement and reinstatement of labor within a given set of firms or industries (Autor et al., 2003; Autor, 2015; Acemoglu and Restrepo, 2018), and work on superstar firms documents rising concentration alongside a falling labor share (Autor et al., 2020). Firm-level evidence on automation shows that adopters expand while competitors shed workers, so that business stealing, rather than the scale effect alone, accounts for much of adopters' growth (Acemoglu et al., 2020; Koch et al., 2021). Our model has no product-market interaction between firms, so it captures the first of these margins and not the second. Hubmer and Restrepo (2021) show that a fixed cost of automation, paid mostly by large firms, can account for the joint dynamics of firm size, labor shares, and capital–labor substitution; our size-selective adoption cost plays the same role. Macroeconomic models in which intangible or information technologies favor large firms (De Ridder, 2024; Aghion et al., 2023) predict that cheaper technology can raise concentration and depress entry, which is the cloud-era pattern of Section 3.

Canonical models of firm dynamics connect entry costs to the firm size distribution (Hopenhayn, 1992; Luttmer, 2007), and the decline in US startups has been traced to demographics and population growth (Karahan et al., 2024; Hopenhayn et al., 2022). On cloud computing, Ewens et al. (2018) show that cheaper experimentation changed how startups are financed, Jin and McElheran (2017) relate cloud adoption to young establishments' survival, and DeStefano et al. (2023) find that cloud adoption raised employment at young firms. Firm-dynamics models have been used much less to ask whether a shock that combines labor-saving technical change with lower entry costs raises or lowers aggregate labor demand; we are not aware of one that does so for AI, which is the gap this paper addresses.

Evidence on AI and labor is accumulating quickly. Administrative data from Denmark show small effects of chatbots on earnings and hours so far (Humlum and Vestergaard, 2025), while US payroll data show falling employment of young workers in the most exposed occupations (Brynjolfsson et al., 2025a). On entry, Dinlersoz et al. (2024) study AI among US startups, Decker and Haltiwanger (2023) analyze the pandemic surge in business formation that precedes our sample's post-period, Bena et al. (2026) find about 20% more startups in AI-exposed industries, with smaller entrants and net job and wage growth among new firms, and Gupta et al. (2026) find that exposed startups cut junior jobs while new firm formation offset the contraction of incumbent startups, leaving aggregate startup employment unchanged. Section 6 positions our business-application evidence against these two studies.

The remainder of the paper is organized as follows. Section 2 presents the framework, Section 3 reviews the cloud era as motivation, Section 4 formalizes the AI shock, Section 5 reports the parameters, the normalization, and the simulations, Section 6 states the empirical signatures and the business-application evidence, Section 7 discusses limitations, ongoing work, and policy, and Section 8 concludes.

# 2 Theoretical Framework

We embed a span-of-control technology (Lucas, 1978) in an entry-and-exit model in the spirit of Hopenhayn (1992), a structure chosen because it links entry and operating costs directly to the firm size distribution. The model is a workhorse in the business dynamism literature (Decker et al., 2014; Haltiwanger et al., 2013), and its extension with random growth yields a Pareto tail of the kind observed in the firm size distribution (Luttmer, 2007; Axtell, 2001). We use it in partial equilibrium: the wage is fixed, product demand is not modeled, and the outcomes are changes in labor demand.

## 2.1 Environment

Time is discrete. A continuum of firms produces a homogeneous good using labor, with idiosyncratic productivity $z$ that follows an AR(1) process in logs,

$$\ln z' = \rho \ln z + (1-\rho)\mu + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma_z^2), \tag{1}$$

with persistence $\rho$ and long-run mean $\mu$. A firm with productivity $z$ that employs $n$ workers produces

$$y = z\, n^{\alpha}, \qquad 0 < \alpha < 1, \tag{2}$$

where $\alpha$ is the span-of-control parameter (decreasing returns to labor), and pays the wage $w$ and a fixed operating cost $c_f$ each period.

## 2.2 Incumbents

Static profit maximization gives labor demand

$$n(z) = \left(\frac{\alpha z}{w}\right)^{\frac{1}{1-\alpha}}, \tag{3}$$

and the value of an incumbent satisfies

$$V(z) = \pi(z) - c_f + \beta \int \max\{0,\, V(z')\}\, dF(z' \mid z), \tag{4}$$

where $\pi(z)$ is static profit before the fixed cost and $\beta$ is the discount factor. Firms exit when $V(z) < 0$, which defines an exit threshold $z^*$. Firms also exit exogenously with probability $\delta$ each period.

## 2.3 Entry

A potential entrant pays a sunk entry cost $c_e$ and then draws initial productivity from a distribution $G(z)$. In general equilibrium with positive entry, free entry would require $\int V(z)\, dG(z) = c_e$, and this condition together with labor-market clearing would pin down the wage. Because we hold the wage fixed, free entry cannot determine the mass of entrants. The simulation instead uses a finite pool of potential entrants whose size responds to the entry cost (Section 4.3).

### 2.4 The fixed-wage stationary distribution and employment accounting

Given the wage $w$, the entry rule, and the exit threshold, the model has an invariant distribution of firms $\mu$, which we compute numerically. We do not close the labor market: labor supply is perfectly elastic at $w$, so any change in labor demand shows up one for one as a change in employment, and no wage or output-price adjustment dampens it. This is a statement about the model, not about the economy. Every employment outcome in this paper is therefore a change in labor demand at a constant wage. Section 7.3 lists the closure the model lacks.

We partition firms by employment into three size classes, the same ones the simulation uses: large firms $\mathcal{L}$ (500 employees or more), mid-sized firms $\mathcal{B}$ (20 to 499 employees), and small firms $\mathcal{S}$ (fewer than 20). Aggregate labor demand is then

$$N = N^{\mathcal{L}} + N^{\mathcal{B}} + N^{\mathcal{S}}, \qquad N^{k} = \int_{z:\, n(z)\in k} n(z)\, d\mu(z), \quad k \in \{\mathcal{L}, \mathcal{B}, \mathcal{S}\}. \tag{5}$$

The 500-employee cutoff matches a standard BDS firm-size class; it is lower than the cutoffs used to define very large firms in the data because the simulated economy has about 180,000 workers and, across five seeds, between zero and three firms with 5,000 or more employees (Section 5.1). European aggregate statistics typically group all firms with 250 or more employees, so European comparisons use that cutoff where necessary. Equation (5) is an accounting identity; the model's content is in how a shock moves firms across the size classes and changes the mass of firms in each one.

### 2.5 The displacement threshold

Whether technological change raises or lowers a firm's labor demand turns on a threshold. We show in Section 4.2 that when AI is a substitute for labor within a CES bundle with elasticity of substitution $\sigma_{\text{ai}}$, AI adoption *increases* a firm's headcount if and only if $\sigma_{\text{ai}}$ is below the firm's scale elasticity $\eta$, the elasticity of its demand for the input bundle with respect to the bundle's unit cost. This is the Hicks–Marshall condition for a fall in the price of a substitute input to raise the demand for labor (Hamermesh, 1993). In the span-of-control model of Equation (2), $\eta = 1/(1-\alpha) \equiv \theta$, so

$$\sigma_{\text{ai}} < \theta = \frac{1}{1-\alpha}. \tag{6}$$

At $\alpha = 0.64$ (the value in Hopenhayn and Rogerson, 1993), $\theta \approx 2.78$. Below this threshold, the scale effect dominates: cheaper effective input lets the firm produce more, and the extra output requires more labor. Above it, the substitution effect dominates: the firm replaces workers with AI services faster than it grows. The threshold is sensitive to $\alpha$, which we discuss in Section 4.2.

The labor–AI substitution elasticity is the central uncertain quantity. The closest available estimate, from Luduvice and Pinheiro (2026), is an elasticity of about 0.89 between AI capital, proxied by computer investment, and a composite of high-skill labor and equipment; that is a different aggregate from our firm-level labor–AI bundle, and it points to complementarity rather than substitution. Our parameter space requires $\sigma_{\text{ai}} > 1$ (gross substitutes) throughout, so complementarity is excluded by construction, and all results below should be read as conditional on AI being a gross substitute for labor at adopting firms. Evidence from earlier automation does not identify the elasticity either: robot adoption raised productivity without clear aggregate employment losses (Graetz and Michaels, 2018), and adopting firms often expanded, which is consistent with a scale effect at adopters but also with business stolen from competitors (Acemoglu et al., 2020; Koch et al., 2021). Survey evidence suggests AI has so far changed headcount at few adopting firms: in

the Census Bureau's Business Trends and Outlook Survey AI supplement (November 2025–January 2026), 2.0% of AI-using firms reported reducing employment because of AI and 2.3% reported increasing it (2.4% and 3.7% employment-weighted) (Bonney et al., 2026). We therefore treat $\sigma_{\text{ai}}$ as unknown and report results as functions of it (Section 5).

# 3 Motivation: The Cloud Computing Era, 2006–2020

## 3.1 Why cloud computing is a natural comparison

Before commercial cloud services, a new technology firm had to buy and operate its own servers before it could test whether its product had demand. The public launch of Amazon Web Services' storage and compute services in 2006 allowed firms to rent computing capacity by the unit instead, converting a fixed, sunk investment into a variable cost. Ewens et al. (2018) exploit this change and show that lower costs of experimentation altered how venture investors financed early-stage firms in the affected sectors, and Jin and McElheran (2017) relate cloud adoption to the capital required to launch a firm and to the survival of young establishments. Using firm-level data, DeStefano et al. (2023) find that cloud adoption raised employment at young firms. In the model of Section 2, this shock maps most naturally into a reduction of the entry cost $c_e$ and the fixed operating cost $c_f$. That is why the cloud era is a natural comparison for the entry-cost component of the AI shock. We use it only as motivation: the facts below describe what happened, and the replication exercise in Section 3.4 shows that the model does not reproduce them through the channel the analogy is about.

## 3.2 Data

For the United States we use the Census Bureau's Business Dynamics Statistics (BDS), which report firm and establishment entry, exit, and employment by firm size and age; we cite them as U.S. Census Bureau (2021), but the figures below come from the release covering 1978–2023 (`data/cloud_era_baseline.md`). The Business Formation Statistics (BFS) track business applications and planned-wage intentions from 2004 (U.S. Census Bureau, 2026a); we use the vintage released on November 12, 2026, with data through October 2026. European comparisons, in Appendix B, use Eurostat business demography statistics and the OECD DynEmp database.

## 3.3 What happened: concentration, not offset

Contrary to the hypothesis that lower fixed costs would proliferate small firms, US employment became more concentrated over the cloud era (U.S. Census Bureau, 2021):

- **Small-firm employment share fell**: from 50.9% (2006) to 47.0% (2019) for firms with fewer than 500 employees, and about 90% of the net rise in employment over the period occurred at firms with 500 or more employees.
- **The startup rate declined**: the employer startup rate fell from 10.3% (2006) to 8.5% (2019), and the employment share of young firms (five years or younger) fell from 13.4% to 10.6%, continuing a secular decline that predates cloud computing (Decker et al., 2016) and that has been linked to slowing labor-force growth (Karahan et al., 2024; Hopenhayn et al., 2022).

- **The largest firms gained share**: firms with 10,000 or more employees raised their share of employment from 26.3% to 29.2%, consistent with the rise of superstar firms documented by Autor et al. (2020).
- **Formation shifted to the nonemployer margin**: business applications rose 33.8% between 2006 and 2019, while applications with planned wages fell 48.1% (U.S. Census Bureau, 2026a).

The most cloud-exposed sector is a partial exception. The startup rate in Information (NAICS 51) recovered from 8.5% in 2010 to 11.3% in 2019, above its 2006 level of 10.6%, whereas professional services, retail, and manufacturing remained below their 2006 levels (U.S. Census Bureau, 2021). These aggregate facts are not evidence that cloud computing caused concentration: demographics, regulation, and the forces that favor large firms in intangible-intensive industries (De Ridder, 2024; Aghion et al., 2023) moved over the same years, and firm-level evidence points to cloud adoption helping young firms grow (DeStefano et al., 2023).

### 3.4 Can the model reproduce the cloud era?

We asked which parameter configurations of the model reproduce the cloud-era pattern of rising concentration and falling dynamism. In this exercise the AI block stands in for cloud and IT: $\sigma$ denotes the elasticity of substitution between labor and the rented IT input inside an adopting firm (the same parameter as $\sigma_{\text{ai}}$ in Section 4.2), and the adopter's unit-cost saving is held at its baseline value when $\sigma$ changes (`data/cloud_replication_scenarios.md`). No single entry-side parameter does it: varying the entry cost, the fixed cost, entrant quality, or the founder pool on its own leaves the small-firm share rising. The best-fitting configuration (Table 1) combines a scale-effect-dominant technology ($\sigma = 2.0$, below $\theta \approx 2.78$ at $\alpha = 0.64$) with entry headwinds: a 20% *rise* in the employer entry cost, a 20% smaller founder pool, and no productivity gain for new entrants.

This result cuts against using the cloud era to discipline the AI shock. The cloud analogy is about a fall in entry costs, yet the configuration that matches the data raises them. With the cloud-like 20% cut in the entry cost instead, the startup rate falls by only 0.33 percentage points against 1.8 in the data (`data/cloud_replication_scenarios.md`, section 5). In the best fit, the technology parameter drives the size distribution and the entry-side headwinds drive the decline in dynamism, so the replication says more about the forces behind declining dynamism (Akcigit and Ates, 2021; Karahan et al., 2024) than about cheaper computing. Because the model's pre-shock levels are far from the data (Section 5.1), the comparison is in changes rather than levels. We therefore do not use the cloud era to set any AI prior.

### 3.5 An empirical design for the cloud era

We have not estimated the causal effect of cloud computing on entry. A design for doing so would compare industries by their ex ante exposure,

$$y_{jt} = a_j + \lambda_t + \sum_{k \neq -1} \beta_k \left( \text{Exposure}_j \times \mathbf{1}\{t = 2006 + k\} \right) + X'_{jt}\gamma + \varepsilon_{jt}, \tag{7}$$

where $j$ indexes industries, $y_{jt}$ is an outcome such as the firm entry rate or the employment share of firms below 500 employees, and $\text{Exposure}_j$ measures pre-2006 reliance on computing infrastructure (for example, IT capital intensity from the BEA). Because the aggregate US startup rate declined over this period (Decker et al., 2016), identification would come from differences across industries relative to a common trend. This estimation is part of the ongoing work listed in Section 7.3.

Table 1: Best-fitting cloud-era configuration: model versus data, 2006–2019

| | |
|---|---|
| *Parameters* | |
| $\sigma$ (labor–IT substitution) | 2.0 (scale effect dominant, $\sigma < \theta$) |
| Employer entry cost | +20% (opposite sign to a cloud cost cut) |
| Entrant productivity shift | 0 |
| Founder pool | −20% |

| Moment (change) | Model | Data |
|---|---|---|
| Employment share of firms < 500 (pp) | −2.3 | −3.9 |
| Startup rate (pp) | −1.6 | −1.8 |
| Young-firm employment share (pp) | −3.1 | −2.8 |

*Notes:* Model values average years 11–15 after the shock (centered on 2019) across 24 seeds, relative to a no-shock counterfactual; $\sigma$ is varied at the baseline unit-cost saving. Data: BDS, 1978–2023 release (U.S. Census Bureau, 2021). Source: `scripts/cloud_replication.py`, `data/cloud_replication_scenarios.md`.

# 4 Formalizing the AI Shock

We model the AI shock as a combination of the displacement channel emphasized in the automation literature and an entry-cost channel.

**Assumption 1** (AI shock)**.** The AI shock has three components, each scaled by a diffusion index $S_t \in [0, 1]$: (i) adopting firms can use AI services, an input that substitutes for labor with elasticity $\sigma_{\text{ai}} > 1$, at a price that falls as AI diffuses; (ii) the entry cost falls from $c_e$ to $\kappa_e c_e$, with $\kappa_e \in (0, 1]$; and (iii) the mean log productivity of entrants rises by $\Delta\mu_e \geq 0$.

Two further objects appear in the exposition but are *not used in any simulation*: the task share $\phi$ of the task-based model in Section 4.1, and a scaling $\kappa_f$ of the fixed operating cost (held at $\kappa_f = 1$ throughout).

## 4.1 Displacement at incumbents: a task-based illustration

To fix ideas, and following Acemoglu and Restrepo (2018), replace the labor input in Equation (2) with a Cobb–Douglas aggregate of a unit continuum of tasks,

$$y = zX^{\alpha}, \qquad \ln X = \int_0^{\phi} \ln a(i)\, di + \int_{\phi}^{1} \ln n(i)\, di, \tag{8}$$

where tasks $i \leq \phi$ are performed by AI services $a(i)$, rented at unit price $r$, and the remaining tasks by labor $n(i)$ at wage $w$. The unit cost of $X$ is $c(\phi) = r^{\phi} w^{1-\phi}$, and firm employment is

$$n(z; \phi) = \frac{1-\phi}{w} (\alpha z)^{\frac{1}{1-\alpha}} c(\phi)^{-\frac{\alpha}{1-\alpha}}, \tag{9}$$

because cost minimization spends the same amount on every task and profit maximization gives $X = (\alpha z/c)^{1/(1-\alpha)}$. Holding $w$ and $r$ fixed, a marginal increase in $\phi$ changes firm employment by

$$\frac{\partial \ln n}{\partial \phi} = -\frac{1}{1-\phi} + \frac{\alpha}{1-\alpha} \ln \frac{w}{r}. \tag{10}$$

The first term is the displacement effect and the second the productivity effect. This formulation fixes the elasticity of substitution between AI and labor at one; the simulations use the CES generalization below instead, and $\phi$ plays no role in them.

### 4.2 CES specification and the displacement threshold

A firm that adopts AI produces $y = zX^{\alpha}$ with a CES bundle of labor $n$ and AI services $a$,

$$X = \left[ n^{\frac{\sigma_{\text{ai}}-1}{\sigma_{\text{ai}}}} + a^{\frac{\sigma_{\text{ai}}-1}{\sigma_{\text{ai}}}} \right]^{\frac{\sigma_{\text{ai}}}{\sigma_{\text{ai}}-1}}, \qquad c(w,r) = \left[ w^{1-\sigma_{\text{ai}}} + r^{1-\sigma_{\text{ai}}} \right]^{\frac{1}{1-\sigma_{\text{ai}}}}, \tag{11}$$

where $r$ is the effective price of AI services and $c$ the unit cost of the bundle. (The simulation code uses this unweighted form, so that a firm without AI, $r \to \infty$, has $c = w$; earlier drafts carried a distribution weight $\omega$ that the code does not use.) Throughout, $\sigma_{\text{ai}} > 1$: AI and labor are gross substitutes, so AI always takes over some work that labor would otherwise do. Whether adoption raises or lowers headcount then depends on whether the scale effect outweighs the substitution effect.

Cost minimization gives $n = (w/c)^{-\sigma_{\text{ai}}} X$. Let $\eta$ denote the firm's scale elasticity, the elasticity of its optimal bundle $X$ with respect to the bundle's unit cost, so that $X \propto c^{-\eta}$ holding $z$ fixed. Under span of control (Equation (2) with a competitive output price), $X = (\alpha z/c)^{1/(1-\alpha)}$ and $\eta = \theta = 1/(1-\alpha)$. Under monopolistic competition with constant returns and CES product demand with elasticity $\varepsilon > 1$, the firm charges a constant markup over marginal cost and $\eta = \varepsilon$; combining decreasing returns $\alpha$ with CES demand gives $\eta = 1/[1 - \alpha(\varepsilon - 1)/\varepsilon]$. In every case, holding $w$ and $z$ fixed, $n \propto c^{\sigma_{\text{ai}}-\eta}$, and since $\partial \ln c / \partial \ln r = s_a$, the cost share of AI in the bundle,

$$\frac{\partial \ln n}{\partial \ln r} = s_a \left( \sigma_{\text{ai}} - \eta \right). \tag{12}$$

This is the Hicks–Marshall decomposition of a cross-price elasticity into a substitution effect ($s_a \sigma_{\text{ai}}$) and a scale effect ($-s_a \eta$) (Hamermesh, 1993). A fall in the price of AI raises labor demand at the firm if and only if the scale effect is larger.

**Proposition 1** (Firm-level displacement threshold)**.** *Let $\sigma_{ai} > 1$, let $\eta$ be the firm's scale elasticity, and hold the wage $w$ and productivity $z$ fixed. A firm that adopts AI at any finite price $r$ employs more workers than it would without AI if $\sigma_{ai} < \eta$, fewer if $\sigma_{ai} > \eta$, and the same number if $\sigma_{ai} = \eta$. The ratio of headcount with and without AI is $(c(w,r)/w)^{\sigma_{ai}-\eta}$, which does not depend on $z$. In the span-of-control model, $\eta = \theta = 1/(1-\alpha)$.*

*Proof.* The relation $n = A\, c^{\sigma_{\text{ai}}-\eta}$, with $A$ independent of $r$, holds at every AI price $r \in (0, \infty]$. A firm without AI is the limit $r \to \infty$, where $c \to w$. With $\sigma_{\text{ai}} > 1$, $c(w,r)$ is strictly increasing in $r$, so $c(w,r) < w$ for any finite $r$. Dividing the two headcounts gives $(c/w)^{\sigma_{\text{ai}}-\eta}$, which is above one if and only if $\sigma_{\text{ai}} < \eta$. The per-period adoption cost $F_{\text{ai}}$ (Section 4.3) affects only whether the firm adopts, not its headcount once it does. □

The proposition is an application of a textbook result, not a new one; we state it because it organizes everything that follows. It is about a single firm at a fixed wage. With business stealing between firms (Acemoglu et al., 2020; Koch et al., 2021), a finite labor-supply elasticity, or downward-sloping market demand, adopters' expansion would come partly at the expense of other firms' labor demand, and the aggregate response would differ.

**The threshold depends on $\alpha$.** Our simulations use $\alpha = 0.64$ (Hopenhayn and Rogerson, 1993), which gives $\theta \approx 2.78$, and draw $\sigma_{\text{ai}}$ from $\mathcal{U}(2,5)$ in the robustness exercise of Section 5.4. Table 2 shows how the threshold, and the share of that prior lying below it, move with $\alpha$. Estimates of returns to scale at the establishment level are often near 0.85 (Atkeson and Kehoe, 2005; Restuccia and Rogerson, 2008); at that value $\theta \approx 6.67$, and every elasticity in the prior support lies in the scale-dominant region. Whether AI-adopting firms are above or below the threshold therefore depends as much on the curvature of the profit function as on the elasticity itself.

Table 2: The displacement threshold $\theta = 1/(1-\alpha)$ and the span-of-control parameter

| $\alpha$ | 0.60 | 0.64 | 0.70 | 0.75 | 0.80 | 0.85 |
|---|---|---|---|---|---|---|
| Threshold $\theta = 1/(1-\alpha)$ | 2.50 | 2.78 | 3.33 | 4.00 | 5.00 | 6.67 |
| Share of $\mathcal{U}(2,5)$ below $\theta$ | 17% | 26% | 44% | 67% | 100% | 100% |

*Notes:* Share $= \min\{1, \max\{0, (\theta - 2)/3\}\}$. The simulations use $\alpha = 0.64$. Source: `scripts/threshold_table.py`, `data/threshold_table.md`.

### 4.3 Lower barriers to entry and size-selective adoption

The parameter $\kappa_e$ captures the part of AI that, like cloud services, turns set-up tasks that previously required dedicated staff or fixed investment into a rented service. Each period a pool of $M_0(c_e/\kappa_e c_e)^{\eta_M}$ potential entrants draws log productivity from $G$, and a draw enters if its expected discounted profit $V(z)$ covers the entry cost $\kappa_e c_e$. Because $V$ is increasing in $z$, this defines a cutoff $z_e$ with $V(z_e) = \kappa_e c_e$, and expected entry is

$$M_e(\kappa_e) = M_0\, \kappa_e^{-\eta_M}\, \big[1 - G(z_e(\kappa_e))\big], \tag{13}$$

which is decreasing in $\kappa_e$ through both terms: a lower entry cost enlarges the pool (pool elasticity $\eta_M = 0.3$) and lowers the cutoff. An incumbent exits when its best available flow profit, net of fixed and AI costs, falls below zero, or exogenously with probability $\delta = 0.05$. The invariant distribution has no closed form, and we characterize it numerically. Because AI is also available to entrants, the shock can reduce the employment of a new firm of given productivity, so more firms need not mean more jobs.

**AI-native entrants.** We allow the log-productivity distribution $G$ from which entrants draw to shift up by $\Delta\mu_e \geq 0$ at full diffusion, capturing the possibility that firms built around AI from the start are more productive than counterfactual entrants.

**Size-selective adoption.** Firms pay a per-period fixed cost $F_{\text{ai}}$ to adopt AI. Because the profit gain from adoption is proportional to $z^\theta$ while $F_{\text{ai}}$ is fixed, only firms above a productivity (and hence size) cutoff adopt, as in models of automation with fixed costs (Hubmer and Restrepo, 2021). The cutoff productivity $\underline{z}$ satisfies

$$\big[K(c_{\text{ai}}) - K(w)\big]\, \underline{z}^{\theta} = F_{\text{ai}}, \tag{14}$$

where $K(c) = (1-\alpha)(\alpha/c)^{\alpha/(1-\alpha)}$ is the profit coefficient at unit cost $c$. This generates the pattern seen in surveys: large incumbents adopt first, concentrating any displacement at the top of the size distribution.

**Sectors and diffusion.** Potential entrants are allocated to four sectors (goods; trade and transport; professional services; information and finance) with shares 0.15, 0.30, 0.35, and 0.20, and AI efficiency in sector $s$ is proportional to $S_t \times e_s$ with assumed relative exposures $e_s = 0.5$, 0.8, 1.2, and 1.5. The effective AI price in sector $s$ is $r_s(t) = \bar{r}/(S_t e_s)$, where $\bar{r}$ is the reference price at full diffusion in an exposure-one sector. Diffusion $S_t$ follows a logistic curve that starts in period 5, with speed 0.5 per year and midpoint 8 years after the start. Table 3 lists all parameters.

### 4.4 The offset ratio

**Definition 1** (Offset ratio)**.** Let $\Delta N^{\mathcal{L}} < 0$ be the change in labor demand at large firms (500 or more employees), relative to the no-AI counterfactual, and let $\Delta N^{\mathcal{B}} + \Delta N^{\mathcal{S}}$ be the change at firms below 500 employees, including entrants. The offset ratio is

$$\Omega = \frac{\Delta N^{\mathcal{B}} + \Delta N^{\mathcal{S}}}{-\Delta N^{\mathcal{L}}}. \tag{15}$$

Firm formation and growth below 500 employees fully offset the loss at large firms when $\Omega \geq 1$, which, when $\Delta N^{\mathcal{L}} < 0$, is the same event as a positive net labor-demand effect $\Delta N > 0$.

Ideally firms would be classified by their size before the shock, which avoids attributing a large firm that shrinks below 500 employees to the growth of smaller firms (see the discussion of size classification in Haltiwanger et al., 2013). The simulation classifies firms by *current* size, so a firm that crosses the 500 cutoff counts as a loss in one class and a gain in the other; the cohort-based version is not yet implemented (Section 7.3). Because $\Omega$ is defined only when large-firm labor demand falls, our main outcome is the net labor-demand effect $\Delta N / N_{\text{cf}}$ relative to the no-AI counterfactual. Proposition 1 does not by itself determine it, which also depends on which firms adopt, on entry and exit, and on the entrant productivity distribution, so we characterize it numerically in Section 5.

## 5 Parameters, Normalization, and Simulation Results

### 5.1 Parameters and the baseline economy

The model is *not calibrated*. Its parameters (Table 3) are illustrative defaults, hand-set to give an entry rate near 7% and a right-skewed firm size distribution; none was estimated, and the simulated method of moments estimator in the code is a stub. Table 4 compares the pre-AI economy, simulated for five seeds, with US data. The model's entry rate is close to the data, but it puts a quarter of employment in firms with 500 or more employees against a little over half in the data, has too few very small firms, and has larger firms on average. Its Pareto tail for firms with 100 or more employees is thinner than a Zipf law would imply, though we have no data target for that cutoff. The simulated economy is also small: about 5,700 firms and 180,000 workers, with about 41 firms of 500 or more employees and between zero and three of 5,000 or more in the final cross-section of each seed. Levels are therefore not comparable with the data, and the quantitative results below are statements about this parameterization, not about the US economy.

### 5.2 Simulation design

For each parameter vector, we simulate the economy for 40 periods (AI diffuses from period 5) and compare it with a no-AI counterfactual run from the same burned-in economy and the same random-number seed, so that each surviving firm receives the same productivity and exit shocks in both runs. Outcomes are averaged over the last 10 periods. The wage is fixed and labor supply perfectly elastic (Section 2.4), so every outcome below is a change in labor demand at a constant wage, not an equilibrium change in employment. The magnitudes reflect this: in the Monte Carlo of Section 5.4 under the original specification, mean output rises by 61% relative to the counterfactual (median 48%), because adopters face neither a rising wage nor a falling output price as they expand (`data/monte_carlo_results.md`). With finite labor supply or downward-sloping product demand, both gains and losses would be smaller. Firms are classified by current size into the three classes of Section 2.4 (fewer than 20, 20 to 499, and 500 or more employees).

Table 3: Model parameters (illustrative defaults, not calibrated)

| Parameter | Description | Value | Basis |
|---|---|---|---|
| *Technology, prices, and costs* | | | |
| $\alpha$ | Span of control (returns to scale) | 0.64 | Hopenhayn and Rogerson (1993); Table 2 |
| $w$ | Wage (numeraire) | 1, fixed | Partial equilibrium |
| $c_f$ | Fixed operating cost | 0.5 | Illustrative |
| $\beta$ | Discount factor; entry-value horizon | 0.96; 40 years | Standard annual |
| *Productivity* | | | |
| $\rho$ | Persistence of log productivity | 0.97 | Illustrative |
| $\sigma_z$ | S.d. of log-productivity shocks | 0.18 | Illustrative |
| $\mu$ | Long-run mean of incumbents' log productivity | 0.95 | Illustrative |
| $G$ | Entrants' log productivity | $\mathcal{N}(0.75, 0.60^2)$ | Below $\mu$, so firms grow with age |
| *Entry and exit* | | | |
| $c_e$ | Entry cost (pre-AI) | 20 | Hand-set: entry rate $\approx 7\%$ |
| $M_0$ | Mean potential entrants per period | 600 | Illustrative |
| $\eta_M$ | Elasticity of entrant pool to entry cost | 0.3 | Illustrative |
| $\delta$ | Exogenous exit rate | 0.05 | Illustrative |
| *Sectors and diffusion* | | | |
| $s_k$ | Sector shares of potential entrants | 0.15, 0.30, 0.35, 0.20 | Assumed |
| $e_k$ | Relative AI exposure by sector | 0.5, 0.8, 1.2, 1.5 | Assumed ordering |
| $S_t$ | Diffusion: logistic, start, speed, midpoint | period 5; 0.5/yr; 8 yrs | Assumed |
| *AI shock: regime map (fixed) and Monte Carlo prior* | | | |
| $\sigma_{\text{ai}}$ | Labor–AI substitution | grid 1.5–5; $\mathcal{U}(2, 5)$ | No estimate; see Section 2.5 |
| $\bar{r}$ | Reference AI price | 1.5, or normalized | Section 5.3 |
| $1 - \kappa_e$ | Entry-cost reduction | 15%; $\mathcal{U}(0, 0.30)$ | Assumed |
| $\Delta\mu_e$ | Entrant productivity shift | grid 0–0.15; $\mathcal{U}(0, 0.15)$ | Assumed |
| $F_{\text{ai}}$ | AI adoption fixed cost | 6; $\mathcal{U}(2, 10)$ | Size-selective adoption |
| $\kappa_f$ | Fixed-cost scaling | 1 | Not varied |

*Notes:* Costs are in units of the annual wage. Each simulation burns in for 150 periods from 1,000 seeded firms and then records 40 periods. Source: `model/parameters.py`, `scripts/regime_map.py`, `scripts/monte_carlo_t6.py`.

### 5.3 Substitutability versus shock size: normalizing the CES

A CES function has three parameters that matter here: the elasticity of substitution, the distribution of weight between inputs, and the efficiency (or price) of each input. Comparing economies that differ in the elasticity alone requires holding the others fixed in an economically meaningful way; otherwise a change in the elasticity also changes the level of output or cost at the starting point, and its effect is confounded with a change in productivity (Klump and de La Grandville, 2000; León-Ledesma et al., 2010). The standard solution is to normalize the function at a baseline point and vary the elasticity holding that point fixed.

Earlier drafts did not normalize. They held the reference AI price $\bar{r}$ fixed at 1.5 for every $\sigma_{\text{ai}}$, so $\sigma_{\text{ai}}$ also set the size of the shock: at the reference point (full diffusion, exposure-one sector, $w = 1$), adoption cuts an adopter's unit cost by 69.7% when $\sigma_{\text{ai}} = 1.5$, by 11.65% when $\sigma_{\text{ai}} = 3.5$, and by 4.4% when $\sigma_{\text{ai}} = 5$ (`data/normalization_cost_saving.md`). A low elasticity was therefore also a very large cost shock, and a high elasticity a small one. The single-firm headcount ratio $(c/w)^{\sigma_{\text{ai}}-\theta}$ of Proposition 1 shows the consequence: under the fixed price it falls from 0.960 at $\sigma_{\text{ai}} = 3$ to 0.900 at $\sigma_{\text{ai}} = 4$ and then stops falling (0.899 at $\sigma_{\text{ai}} = 4.5$, 0.905 at $\sigma_{\text{ai}} = 5$), because the shrinking cost

Table 4: Pre-AI economy: model versus data

| Moment | Model | Data | Data source |
|---|---|---|---|
| Entry (startup) rate | 7.3% | 8.5% | BDS, 2019 |
| Exit rate | 7.3% | not sourced | – |
| Employment share, firms with 500+ employees | 25.2% | 53.0% | BDS, 2019 |
| Share of firms with fewer than 20 employees | 73.5% | 87.9% | BDS, 2019 |
| Mean firm size (employees) | 31.4 | 24.7 | BDS, 2019 |
| Pareto tail exponent, firms with 100+ employees | 1.28 | not sourced | – |

*Notes:* Model: default parameters, AI off, mean over five seeds of the average over 40 recorded periods; tail exponent by the Hill estimator on the final cross-section. Across seeds the 500+ share ranges from 21% to 28%. Data: BDS 1978–2023 release, as tabulated in `data/cloud_era_baseline.md`. The placeholder targets in `model/calibration.py` (exit rate 8%, tail exponent 1.06 from Axtell, 2001) are unverified, and the Zipf exponent is fitted to the whole distribution rather than to firms with 100+ employees. Source: `scripts/baseline_moments.py`, `data/baseline_moments.md`.

saving offsets the stronger substitution.

The `cost_saving` normalization removes this confound. For each $\sigma_{\text{ai}}$ it sets the reference price $\bar{r}(\sigma_{\text{ai}})$ so that adoption at the reference point lowers the unit cost by the same 11.65%, the saving the original specification delivers at $\sigma_{\text{ai}} = 3.5$:

$$c(1, \bar{r}(\sigma_{\text{ai}})) = 1 - 0.1165 \quad \Longleftrightarrow \quad \bar{r}(\sigma_{\text{ai}}) = \left[(1 - 0.1165)^{1-\sigma_{\text{ai}}} - 1\right]^{\frac{1}{1-\sigma_{\text{ai}}}}. \tag{16}$$

The two specifications therefore coincide at $\sigma_{\text{ai}} = 3.5$. Under the normalization, $\sigma_{\text{ai}}$ changes only how easily AI replaces labor, and the single-firm headcount ratio $0.8835^{\sigma_{\text{ai}}-\theta}$ falls monotonically, from 1.171 at $\sigma_{\text{ai}} = 1.5$ to 0.759 at $\sigma_{\text{ai}} = 5$. Away from the reference point (other sectors, partial diffusion) the saving still varies with $\sigma_{\text{ai}}$, so the normalization fixes the size of the shock at one point rather than everywhere.

**The regime map.** Figure 1 is the paper's central quantitative result. It reports the mean net labor-demand effect, relative to the no-AI counterfactual, on a grid of $\sigma_{\text{ai}}$ and the entrant productivity shift $\Delta\mu_e$, holding the entry-cost cut at 15% and the adoption fixed cost at 6, with 10 seeds per cell. These are comparative statics of the model, not probabilities.

**Result 1** (Regime map, numerical)**.** *In the normalized specification (Figure 1, right):*

*(i) For every entrant productivity shift on the grid, the mean net labor-demand effect declines monotonically in $\sigma_{ai}$. With no entrant shift it falls from +15.3% at $\sigma_{ai} = 1.5$ to −21.1% at $\sigma_{ai} = 5$, crossing zero near $\sigma_{ai} = 3$ (+0.1%, s.e. 1.7%), close to the single-firm threshold $\theta \approx 2.78$.*

*(ii) Offset at high substitution requires more productive entrants, and the required shift rises with $\sigma_{ai}$: the effect changes sign between $\sigma_{ai} = 3.5$ and 4 with $\Delta\mu_e = 0.05$, between 4 and 4.5 with $\Delta\mu_e = 0.10$, and between 4.5 and 5 with $\Delta\mu_e = 0.15$.*

*(iii) At $\sigma_{ai} = 5$ the effect is negative for every entrant shift on the grid, from −21.1% ($\Delta\mu_e = 0$) to −3.3% ($\Delta\mu_e = 0.15$).*

The original specification (Figure 1, left) tells a different story, and the difference is the confound described above. At $\sigma_{\text{ai}} = 1.5$ with no entrant shift, the net effect is +373.9%, which reflects a 70% cost cut rather than easy substitution. At high $\sigma_{\text{ai}}$ the effect flattens: with no entrant shift it is −6.8%, −7.2%, and −7.1% at $\sigma_{\text{ai}} = 4$, 4.5, and 5. That plateau, which earlier drafts interpreted

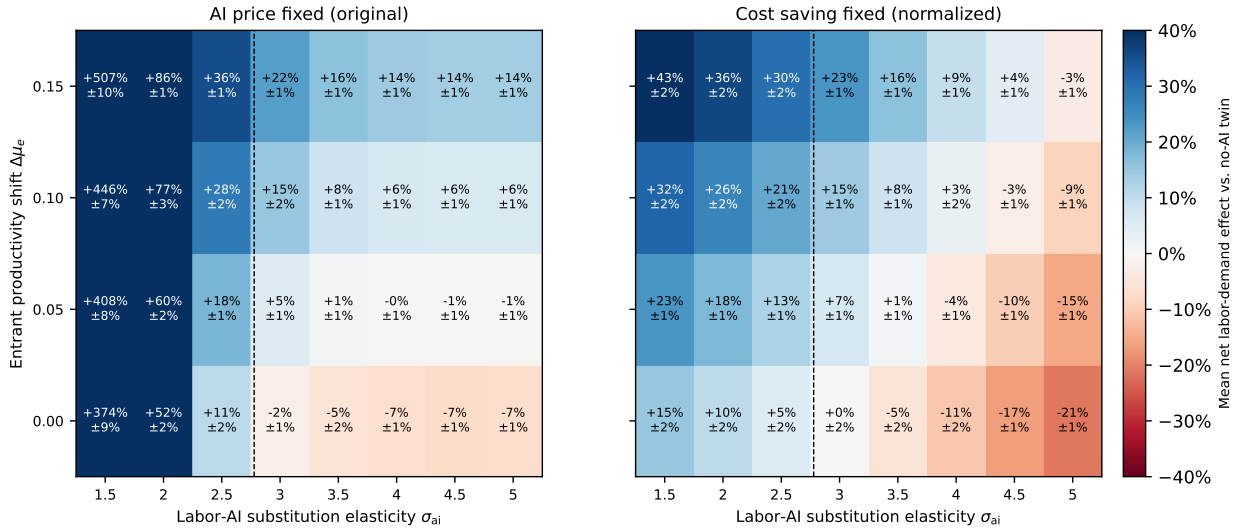



Figure 1: Regime map: mean net labor-demand effect of AI by labor–AI substitution elasticity and entrant productivity shift. Each cell is the mean, over 10 seeds, of the net labor-demand effect relative to the no-AI counterfactual at a fixed wage, with its seed standard error. Left: original specification, AI price fixed at 1.5 for every $\sigma_{ai}$, so a low $\sigma_{ai}$ is also a large cost shock (the color scale is capped at ±40%; cells at $\sigma_{ai} \leq 2$ exceed it). Right: normalized specification, AI price set so that the adopter's unit-cost saving at the reference point is 11.65% for every $\sigma_{ai}$. The dashed line is $\theta = 1/(1-\alpha) \approx 2.78$ at $\alpha = 0.64$. Fixed: entry-cost cut 15%, adoption fixed cost 6. Source: `scripts/regime_map.py`, `data/regime_map.md`.

as offset "levelling off" rather than collapsing at high substitution, is an artifact of holding the AI price fixed: as $\sigma_{ai}$ rises, the shrinking cost advantage of AI reduces adoption and so masks the stronger substitution. With the cost saving held fixed, the plateau disappears, and at $\sigma_{ai} = 5$ the net effect is $-21.1\%$ rather than $-7.1\%$ with no entrant shift.

The headline is therefore conditional and qualitative: *offset fails when substitution is high and AI-native entrants are no more productive than earlier entrants.* Where on the map the economy lies is unknown. The threshold that separates the regions moves with $\alpha$ (Table 2), the only available estimate of a related elasticity is below one (Luduvice and Pinheiro, 2026), and nothing in the paper measures the entrant productivity shift. A companion note derives the industry-level offset condition analytically and shows the same trade-off between substitution and entrant quality.

### 5.4 Robustness: Monte Carlo over an assumed prior

Earlier drafts led with a Monte Carlo over priors for the four AI parameters and reported shares of draws as probabilities. They are not probabilities about the world. The priors in Table 3 are assumptions, chosen to straddle the threshold at $\alpha = 0.64$ rather than to match data, so a share of draws restates the prior as much as it measures the model. We keep the Monte Carlo only as a robustness check on what drives variation within the assumed parameter space, and we report shares of draws under the assumed prior without confidence intervals. Each run uses 2,000 draws with one simulation seed per draw and the same seed (2026) in both specifications.

Table 5: Monte Carlo over the assumed prior: original versus normalized CES

| | Original (AI price fixed) | | Normalized (cost saving fixed) | |
|---|---|---|---|---|
| *First-order variance share of the net labor-demand effect* | | | | |
| $\sigma_{\text{ai}}$ (substitution) | 0.714 | | 0.618 | |
| $\Delta\mu_e$ (entrant productivity shift) | 0.187 | | 0.250 | |
| $1-\kappa_e$ (entry-cost reduction) | 0.039 | | 0.038 | |
| $F_{\text{ai}}$ (adoption fixed cost) | 0.006 | | 0.014 | |
| $\sigma_{\text{ai}}$ bin | Mean effect | Share $>0$ | Mean effect | Share $>0$ |
| $[2.0, 2.5)$ | +42.2% | 100% | +21.6% | 99% |
| $[2.5, 3.0)$ | +17.3% | 98% | +16.3% | 98% |
| $[3.0, 3.5)$ | +8.8% | 82% | +10.0% | 85% |
| $[3.5, 4.0)$ | +6.3% | 73% | +4.2% | 65% |
| $[4.0, 4.5)$ | +4.4% | 68% | −2.9% | 33% |
| $[4.5, 5.0)$ | +4.6% | 68% | −8.5% | 15% |

*Notes:* 2,000 draws per specification from the priors in Table 3, seed 2026. "Share $>0$" is the share of draws in the bin with a positive net labor-demand effect, under the assumed prior. Variance shares are not bias-adjusted; the bias-adjusted shares for the original run in `data/monte_carlo_results.md` are 0.711, 0.179, 0.030, and −0.004. Source: `scripts/monte_carlo_normalized.py`, `data/monte_carlo_normalized_results.md`.

**Variance decomposition, original versus normalized.** Table 5 reports first-order variance shares of the net labor-demand effect, $\text{Var}(E[Y \mid X_i])/\text{Var}(Y)$ estimated from 20 quantile bins, in the two specifications. In the original specification, $\sigma_{\text{ai}}$ accounts for 0.714 of the variance; with the cost saving held fixed it accounts for 0.618, and the entrant productivity shift rises from 0.187 to 0.250. Some of $\sigma_{\text{ai}}$'s apparent importance therefore came from its role in scaling the shock, and entrant quality matters more once that role is removed. The entry-cost reduction accounts for about 0.04 of the variance in both specifications. First-order shares sum to less than one (0.946 and 0.920) because the parameters interact. The interactions matter more for the sign of the effect than for its level: whether the net effect is positive depends on $\sigma_{\text{ai}}$ and $\Delta\mu_e$ jointly, as the regime map shows, so a parameter with a modest first-order share can still decide the outcome in part of the space.

The by-$\sigma_{\text{ai}}$ profile in Table 5 repeats the regime-map finding over the prior. In the original specification, the share of draws with a positive net effect levels off at 68% for $\sigma_{\text{ai}} \geq 4$; in the normalized specification it falls to 33% for $\sigma_{\text{ai}} \in [4, 4.5)$ and 15% for $\sigma_{\text{ai}} \in [4.5, 5)$, and the mean effect turns negative above $\sigma_{\text{ai}} = 4$. Large-firm labor demand falls in 51% of normalized draws against 44% in the original.

**Other prior choices (original specification).** Two further checks were run on the original specification only. Among draws in which large-firm labor demand falls, the share with a positive net effect is 59.2% when the AI price is fixed and 44.5% when it is drawn from a lognormal prior with median 1.5 and log standard deviation 0.4 (`data/monte_carlo_aicost_results.md`); reweighting the draws to alternative priors on $\sigma_{\text{ai}}$ over the same support moves it between 55.7% and 63.3%, and moves the unconditional share of draws with a positive net effect between 69.7% and 89.0% (`data/prior_sensitivity.md`). These are shares of draws under assumed priors, and the spread across priors is the point: they are not estimates of how likely offset is. Both checks predate the normalization and have not been rerun under it.

**Simulation noise.** Each Monte Carlo draw uses a single simulation seed. Sixty replications at fixed parameters ($\sigma_{\text{ai}} = 3.5$, a 20% entry-cost cut, $\Delta\mu_e = 0.05$, $F_{\text{ai}} = 5$, original specification) give a net effect with standard deviation 4.2%, about 6% of the total variance across draws; for $\sigma_{\text{ai}} \geq 3.25$ noise is about a quarter of the variance, because the 500+ class holds few firms (`data/monte_carlo_results.md`). A multi-seed Monte Carlo is part of the ongoing work.

**Implications for monitoring.** The regime map points to two quantities to monitor as AI diffuses:

1. **Employment at AI-adopting firms relative to comparable non-adopters**, which bears on whether adopters are in the scale-dominant or substitution-dominant region. The BTOS AI supplement, in which about 2% of AI-using firms reported AI-related employment reductions (Bonney et al., 2026), records realized cuts at an early stage of diffusion; it does not measure $\sigma_{\text{ai}}$ or $\alpha$.
2. **Productivity and employment at AI-native entrants**, that is, whether firms created with AI are more productive than counterfactual entrants and how many workers they employ per unit of output.

## 6 Empirical Signatures

The model implies a set of predictions that distinguish two hypotheses. Under *pure displacement*, AI lowers labor demand at large incumbents with no offsetting response at the entry margin. Under *offsetting entry*, the same shock also lowers the cost of starting and running a firm, so entry and employment below 500 employees rise in AI-exposed industries. Each prediction below is stated relative to less-exposed industries in the same country and year.

**Prediction 1** (Incumbent headcount)**.** Employment at firms that were large before the shock falls in AI-exposed industries, provided adopters are in the substitution-dominant region ($\sigma_{\text{ai}}$ above the scale elasticity). This prediction holds under both hypotheses and serves as a check that the exposure measure captures the displacement channel.

**Prediction 2** (Entry)**.** Firm entry rates rise in AI-exposed industries relative to their pre-shock trend. Under pure displacement, entry does not respond, or falls if displaced workers face weaker demand for their skills.

**Prediction 3** (Entrant size)**.** Average employment at entry falls, because AI is also available to entrants, while the number of entrants rises.

**Prediction 4** (Employment below 500 employees)**.** Employment at firms below 500 employees, classified by pre-shock size and including entrants, rises in AI-exposed industries. The offset ratio $\Omega$ in Equation (15) is the ratio of this response to the response in Prediction 1.

**Prediction 5** (Capital required at entry)**.** The amount of external capital raised before a firm's first product falls in AI-exposed sectors, by analogy with the cloud era (Ewens et al., 2018).

**Prediction 6** (Transatlantic differences)**.** The employment response below 500 employees (250 in European data) is larger in the United States than in Europe, and the gap appears mainly in the post-entry growth of AI-exposed entrants rather than in entry rates. Europe's employer start-up rate is about half the US rate, its employment protection is about three times as strict,

Table 6: Empirical signatures and candidate data sources

| Prediction | United States | Europe |
|---|---|---|
| 1. Incumbent headcount | BDS; BTOS AI suppl. (2025–26: 2% cut) | Community Survey on ICT |
| 2. Entry | BFS (2025: +62.8% vs. 2019) | Eurostat business demography |
| 3. Entrant size | BDS; BFS (WBA share 2026: 6.8%) | OECD DynEmp |
| 4. Employment below 500 | Business Dynamics Statistics | Eurostat SBS |
| 5. Capital at entry | PitchBook, Crunchbase | Dealroom |
| 6. Transatlantic gap | Harmonized via OECD DynEmp | |

BFS figures from U.S. Census Bureau (2026a), November 12, 2026 vintage: business applications in 2025 relative to 2019 (unadjusted annual totals), and applications with planned wages (WBA) as a share of all applications, January–October 2026 (`data/bfs_revision_numbers.md`). BTOS figure from Bonney et al. (2026).

and its private AI investment is roughly a tenth of the US level; formal start-up registration costs, by contrast, are not systematically higher (Eurostat, 2026a; OECD, 2020, 2024; Stanford Institute for Human-Centered Artificial Intelligence, 2026). Appendix B reports a stylized European parameterization of the model.

Table 6 maps each prediction to candidate US and European data sources.

**AI exposure measure.** This draft uses a coarse, ex ante classification rather than a continuous exposure score. In the business-application evidence, the exposed sectors are Information (NAICS 51) and Professional, scientific, and technical services (NAICS 54), which rank among the most exposed on the task-based measures of Eloundou et al. (2024) and Felten et al. (2021) and have among the highest AI use rates in BTOS. Finance and insurance (NAICS 52) is also highly exposed on those measures, and we report what happens when it is added. The simulation uses assumed relative exposures (Table 3) whose ordering is consistent with AI use rates by sector in BTOS and Eurostat but which are not estimated from them. Mapping occupation-level scores to NAICS and NACE industries, which would give a continuous treatment intensity, is part of the ongoing work.

### 6.1 Business applications: descriptive evidence

A first look at Prediction 2 uses the Census Business Formation Statistics (U.S. Census Bureau, 2026a), November 12, 2026 vintage: seasonally adjusted monthly applications for 19 NAICS sectors through October 2026. This vintage adds October 2026 and revises earlier months; for example, applications in January–September 2026 are now 15.0% above the same months of 2025, against 15.3% in the previous vintage. The annual outcome is the log of mean monthly applications (2026 covers January–October). We estimate a two-way fixed-effects event study with Information and Professional services as the exposed group, the 15 sectors outside the exposed and placebo groups as controls, and 2022 as the base year, and we compare the exposed group's gap with that of every other pair of sectors. With two exposed sectors, conventional clustered errors over-reject (Conley and Taber, 2011), so we treat the permutation ranks as the primary summary. None of what follows identifies an effect of AI; it describes when and where applications diverged.

**Pre-trends and timing.** Before 2023 the exposed sectors track the controls closely. Their differential trend over 2010–2022 is −0.2% per year (s.e. 1.0%), every 2015–2021 event coefficient lies between 0.01 and 0.08 log points, and a fictitious shock dated 2016, estimated on pre-COVID data, yields −0.001 (s.e. 0.053). The gap is −0.03 in 2023 and 0.00 in 2024. In the monthly series it stays above 0.05 log points from March 2025 onward and widens to 0.34 by September 2026 and

0.44 in October 2026 (Figure 2); October is the newest month and the most likely to be revised. The 2026 event coefficient is 0.30 (s.e. 0.06). Measured against all 17 other sectors, the 2026 gap is 0.288 log points and ranks 10th of the 171 sector pairs that could have been labelled exposed (one-sided permutation $p = 0.058$), whereas the same statistic for 2015–2021 never ranks above 41st. A pooled 2023–2026 difference-in-differences is small (0.07, s.e. 0.06; the corresponding permutation statistic ranks 50th, $p = 0.29$), because 2023 and 2024 show nothing. Dating the divergence to 2025, when capable coding and agent tools diffused, fits the data, but that date was chosen after the data were observed.

**What drives the permutation rank.** All nine sector pairs with a larger 2026 gap than Information plus Professional services include Management of companies (NAICS 55), whose applications more than doubled between 2022 and 2026 (`data/bfs_sector_league_BA_BA.csv`). The rank of 10 is therefore a statement about one small sector as much as about the exposed pair: if NAICS 55 is dropped from the comparison set, the exposed pair ranks first of 153 pairs ($p = 0.007$). Because that exclusion is chosen after seeing the data, we do not use it as the primary result.

**Which sectors count as exposed.** Finance and insurance is highly exposed on task-based measures, but its applications fell relative to other sectors: its 2026 gap relative to 2022, against the 14 controls that exclude the placebo and exposed sectors, is −0.078 (s.e. 0.062), and −0.090 when every year uses January–October only. Adding Finance to the exposed group roughly halves the estimate. The 2026 gap of Information, Finance, and Professional services against all other sectors is 0.160, 0.55 times the two-sector gap, and ranks 121st of the 969 possible sector triples ($p = 0.125$); the event-study coefficient falls to 0.17 (s.e. 0.13). The divergence is therefore a feature of two sectors, not of AI exposure as such.

**Weighting.** The group gap is an unweighted mean of the two sectors' log applications, so Information and Professional services count equally although Professional services accounted for 86% of the pair's applications in 2022. Weighting by applications instead (the log of the pair's summed applications against the log of the controls' summed applications) gives a 2026 gap of 0.32 against 0.30 unweighted, so equal weighting does not drive the result.

**Who is applying.** The divergence is concentrated in applications with a low propensity to hire. For high-propensity applications the 2026 coefficient is 0.11 (s.e. 0.12), ranking 46th of 171, and for Information alone it is 0.005 (s.e. 0.09). Across all sectors, applications in January–October 2026 were 10.9% above the same months of 2025, high-propensity applications 3.6% above, and applications with planned wages 16.5% below; planned-wage applications were 6.8% of all applications (4.7% in Information) (`data/bfs_revision_numbers.md`). This composition fits Prediction 3, more but smaller entrants. It fits a rival reading equally well: workers displaced from, or not hired into, exposed occupations registering as contractors or sole proprietors. Falling employment of young workers in exposed occupations (Brynjolfsson et al., 2025a) and exposed startups cutting junior jobs (Gupta et al., 2026) are consistent with that reading, and business applications cannot separate the two. They also cannot show employer entry.

**Placebos and the 2020 jump.** Construction, a low-exposure sector, stays close to zero after 2022 (event coefficients between −0.04 and 0.00 in 2023–2026). Retail is not a clean placebo. It carries a +3.8% annual pre-trend from the e-commerce wave, and its event coefficients are 0.24 in 2025 and 0.16 (s.e. 0.06) in 2026, more than half the exposed group's 2026 coefficient, in a sector

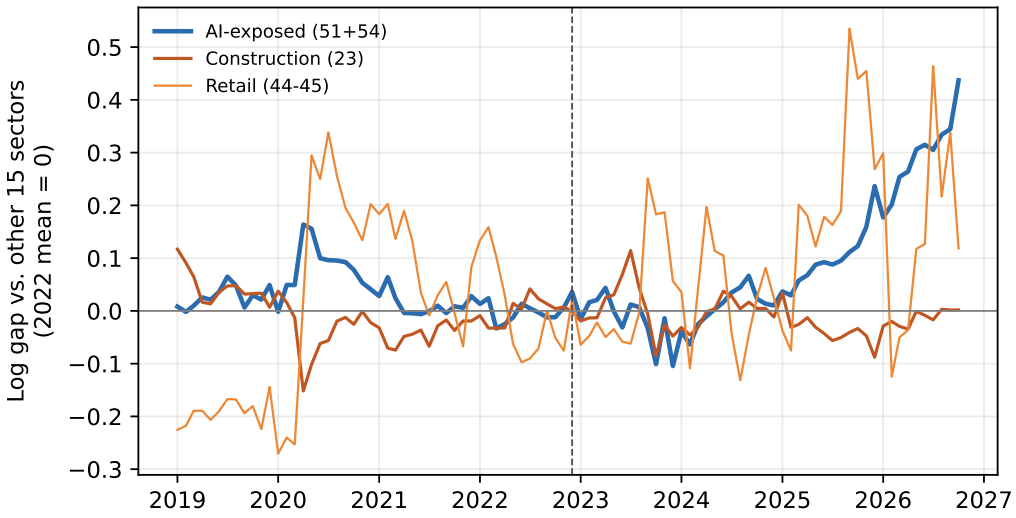



Figure 2: Business applications in AI-exposed sectors relative to other sectors. Log gap between seasonally adjusted monthly applications in each group and the mean of the 15 other NAICS sectors, normalized to the 2022 mean. AI-exposed is the unweighted mean of Information (51) and Professional services (54). Dashed line: December 2022. The exposed-sector gap also rose in 2020 (peak April 2020) and reverted. *Source: Census BFS, November 12, 2026 vintage, January 2019–October 2026;* `scripts/bfs_causal_analysis.py`.

where AI exposure is low. Large sector-specific movements in applications are therefore not rare. The exposed-sector gap itself jumped in 2020: the annual coefficient is 0.08 and the permutation gap ranks 41st of 171, and the monthly gap peaked at 0.16 in April 2020 before reverting, during the pandemic surge in business formation (Decker and Haltiwanger, 2023). The 2026 gap is larger and has persisted longer, but the 2020 episode shows that a gap of this kind can open for reasons unrelated to AI.

**Relation to firm-level evidence.** Bena et al. (2026) find about 20% more startups in AI-exposed industries, with smaller entrants but net job and wage growth among new firms. Our application data agree on the first two points (more formation in exposed sectors, mostly at the low-propensity margin) but cannot speak to the third. Gupta et al. (2026) find that exposed startups cut junior jobs while new firm formation offset the contraction of incumbent startups, leaving aggregate startup employment unchanged, which is closer to the employment-neutral reallocation of the regime map's middle region than to either pure displacement or strong offset. Dinlersoz et al. (2024) study AI among US startups in Census data, which is the natural route to linking applications to later employer outcomes. None of these studies, nor ours, measures labor demand at firms that were large before the shock, which is the denominator of $\Omega$. Data, code, and tables are in `scripts/bfs_causal_analysis.py` (with its CSV outputs in `data/`) and `scripts/bfs_revision_numbers.py` (`data/bfs_revision_numbers.md`).

Table 7: Pre-trend, placebo and permutation tests: log business applications

| | Pre-trend 2010–22 (%/yr) | Fake shock 2016 | Event coef. 2026 | Rank of 2026 gap |
|---|---|---|---|---|
| *All applications* | | | | |
| Info. + Prof. (51, 54) | −0.23 (1.02) | −0.001 (0.053) | 0.296 (0.058) | 10 of 171 |
| Info. + Finance + Prof. (51, 52, 54) | −0.04 (1.06) | 0.040 (0.062) | 0.168 (0.129) | 121 of 969 |
| Placebo: Construction (23) | 2.30 | 0.189 | −0.013 | – |
| Placebo: Retail (44–45) | 3.84 | 0.094 | 0.160 | – |
| *High-propensity applications* | | | | |
| Info. + Prof. (51, 54) | −0.18 (1.02) | 0.019 (0.058) | 0.105 (0.120) | 46 of 171 |

Two-way (sector, year) fixed effects; controls are the 15 NAICS sectors outside the exposed and placebo groups (14 when Finance is in the exposed group); base year 2022; 2026 is January–October. Standard errors clustered by sector in parentheses (reported for the exposed groups only). Fake shock: 2016–2019 vs. 2010–2015. Rank: position of the group's 2026 gap relative to 2022, measured against all other sectors, among all sector pairs ($\binom{19}{2} = 171$) or triples ($\binom{19}{3} = 969$), 1 = largest. All nine pairs ranked above the exposed pair include Management of companies (55). *Source: Census BFS, November 12, 2026 vintage;* `data/bfs_causal_results.csv`, `data/bfs_perm_by_year_BA_*.csv`, `data/bfs_revision_numbers.md`.

# 7 Discussion

## 7.1 Employment-light AI firms

If AI-induced formation offsets displacement, it must do so mainly through *application* firms that build products on AI models, rather than through the frontier labs and cloud providers that supply them, whose growth is capital-intensive and whose employment lands largely at old, large firms. The application firms that have scaled fastest are strikingly employment-light. Cursor had roughly $3.75 million of annualized revenue per employee in 2026 and Lovable about $2.7 million, against about $0.24 million at Snowflake at the end of its IPO fiscal year (Table 8 in Appendix A). Two cautions apply. These comparisons set run-rate revenue at a point of very fast growth against a full year of reported revenue. And revenue is not value added: application firms buy model access from AI providers, so part of their revenue pays for an input produced, and employing people, elsewhere. A handful of firms is not systematic evidence. The systematic evidence points the same way, however: AI-native startups are 12–25% smaller in headcount than comparable firms (Kim and Koning, 2026), and Bena et al. (2026) find smaller entrants in exposed industries. In the model, smaller entrants are what one expects when AI is also available to new firms (Prediction 3); whether they add up to offset depends on how many there are and how fast they grow.

## 7.2 More firms, not more employers

Business applications in January–October 2026 were 10.9% above the same months of 2025, and 36.9% above in Information (U.S. Census Bureau, 2026a). Applications with planned wages, a proxy for intent to become an employer, fell 16.5% over the same period and fell in every one of the 19 sectors. In Information, planned-wage applications were 12.3% of all applications in 2019 and 4.7% in January–October 2026 (`data/bfs_revision_numbers.md`). Formation is rising at the firm-count margin but not at the employer margin, as in the event-study evidence of Section 6.1. Because the decline in planned-wage applications appears in every sector, it is not specific to AI, and it continues a cloud-era trend, when such applications fell 48.1% between 2006 and 2019 (Section 3.3). The pattern is consistent with more but smaller firms, and also with the contractor-registration reading discussed in Section 6.1.

### 7.3 Limitations and ongoing work

- **No equilibrium closure (ongoing).** The model holds the wage fixed and has no product-market interaction, so its outcomes are changes in labor demand, not employment. Closing it requires a finite labor-supply elasticity, downward-sloping product demand, and business stealing between adopters and their competitors, which firm-level studies of automation find to be quantitatively important (Acemoglu et al., 2020; Koch et al., 2021). Each would damp both gains and losses, and business stealing could make the aggregate effect of adoption smaller than the firm-level effect in Proposition 1.

- **Not calibrated (ongoing).** The parameters are illustrative, and the pre-AI economy misses the data on the large-firm employment share by more than half (Table 4). A simulated-method-of-moments calibration to verified BDS and SUSB targets, with $\alpha$ estimated or varied rather than fixed at 0.64, is the next step.

- **Size classes and cohorts (ongoing).** The simulation classifies firms by current size at a 500-employee cutoff; the cohort-based offset ratio of Definition 1 and alternative cutoffs have not been implemented.

- **Monte Carlo design (ongoing).** Each draw uses one simulation seed, and the AI-price and prior-reweighting checks have been run only on the unnormalized specification. A multi-seed Monte Carlo under the normalized CES is pending.

- **Cloud-era estimation (ongoing).** The exposure-based design of Section 3.5 has not been estimated, so the cloud era remains motivation.

- **Continuous exposure (ongoing).** The business-application evidence uses a binary classification of two sectors and is sensitive to that choice; a continuous exposure measure mapped to industries, and outcomes for employer entry rather than applications, are needed.

- **Elasticity support.** The parameter space requires $\sigma_{\text{ai}} > 1$, so complementarity between AI and labor, which the only related estimate suggests (Luduvice and Pinheiro, 2026), is excluded by construction.

- **Sector heterogeneity.** Sector exposures are assumed and fixed; offset could fail in highly exposed sectors and hold in others, and the model does not endogenize reallocation across sectors.

- **Job counts versus job quality.** The offset ratio counts jobs and is silent on wages, job security, and the matching of displaced workers to new firms.

### 7.4 Policy implications

In this model, cutting the cost of entry changes where employment sits more than how much of it there is: across the specifications, the entry-cost reduction accounts for about 4% of the variance in the net labor-demand effect (Table 5). What decides the sign is whether adopters are in the substitution-dominant region and whether AI-native entrants are more productive (Result 1). Three implications follow, each conditional on the model's partial-equilibrium structure. First, policies that only lower the cost of starting a business are unlikely to produce offset on their own if AI is strongly labor-substituting at adopters. Second, the quantities worth measuring are labor demand at AI-adopting firms relative to comparable non-adopters and the productivity and employment of

AI-native entrants; early survey evidence (Bonney et al., 2026) records few AI-related cuts so far but does not measure the substitution elasticity. Third, the conditions for scale-up (finance, and access to AI services and compute on competitive terms) matter more for offset in this framework than the cost of registration, which is also the conclusion of the stylized European comparison in Appendix B.

# 8 Conclusion

This paper asks under what conditions AI-induced firm formation can offset the labor that large incumbents shed as they adopt AI. We model the AI shock as labor-substituting technology at adopters, lower entry costs, and possibly more productive entrants, in a partial-equilibrium firm-dynamics model with a fixed wage. Every result is a change in labor demand, not in equilibrium employment, and the model is not calibrated.

**Main findings.** At a single firm, whether AI raises or lowers headcount is the Hicks–Marshall comparison of the labor–AI substitution elasticity with the scale elasticity. In the span-of-control model the threshold is $1/(1-\alpha)$, which is about 2.78 at $\alpha = 0.64$ but rises steeply with $\alpha$, so the regime an economy is in depends on returns to scale as much as on the elasticity. Once the size of the AI cost shock is held fixed as the elasticity varies, the net labor-demand effect declines monotonically in the elasticity. Offset fails when substitution is high and AI-native entrants are no more productive than earlier entrants; it can survive high substitution only if entrants are substantially more productive. Cutting entry costs moves aggregate labor demand little in any specification. The high-substitution plateau reported in earlier drafts was an artifact of holding the AI price fixed.

In Census business applications, Information and Professional services pulled away from other sectors from March 2025, with no earlier pre-trend, mostly among applicants unlikely to hire, while applications with planned wages fell in every sector. That pattern fits more but smaller entrants, and it also fits displaced workers registering as contractors. It depends on the choice of exposed sectors (adding Finance roughly halves it) and on one small comparison sector, and we give it no causal reading.

**What to watch.** The regime map identifies two quantities: labor demand at AI-adopting firms relative to comparable non-adopters, which bears on whether adopters are above or below the threshold, and the productivity and employment of AI-native entrants. Establishment-level surveys such as the Business Trends and Outlook Survey and the Community Survey on ICT Usage, linked to administrative data on firm entry and growth, can track both as AI diffuses.

**What is not done.** The model does not yet close the labor and product markets, is not calibrated, classifies firms by current rather than pre-shock size, and runs its Monte Carlo with one seed per draw; the cloud-era exposure design and the continuous-exposure business-formation analysis have not been estimated (Section 7.3). Until they are, the paper's contribution is a framework and a conditional map of where offset holds and fails, not an estimate of whether it will.

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# A Company Evidence on Employment Intensity

Table 8 compares valuation, revenue, and headcount at AI infrastructure firms, AI application firms, and cloud-era software firms. It is illustrative. Revenue is not value added: application firms pay model providers for access, so their revenue per employee overstates the value they add per employee, and run-rate figures at fast-growing firms are not comparable with a full year of reported revenue. The systematic evidence that AI-native firms are smaller is Kim and Koning (2026).

Table 8: Valuation and revenue per employee: AI versus cloud-era firms

| Company | Layer / Era | Valuation | Revenue | Employees | Rev./Emp. |
|---|---|---|---|---|---|
| *AI infrastructure (labs, neoclouds)* | | | | | |
| Anthropic (S-1, Oct 2026) | Lab | ∼$1T | ≈$35B TTM (derived) | 5,300 | >$6.6M |
| CoreWeave (2025) | Neocloud | $27B | $5.1B | 2,189 | $2.3M |
| *AI applications (API consumers)* | | | | | |
| Cursor (2026) | Application | $60B | $3B ARR | ∼800 | $3.75M |
| Lovable (Feb 2026) | Application | $13.3B | $400M ARR | 146 | $2.7M |
| Vercel (2025–26) | Application | $9.3B | $340M ARR | 550 | $0.6M |
| *Cloud-era SaaS (baseline)* | | | | | |
| Databricks (2021) | Cloud SaaS | $38B | $0.8B ARR | n/a | n/a |
| Snowflake (FY2021) | Cloud SaaS | $33.3B | $0.59B | 2,495 | $0.24M |

*Sources:* Anthropic S-1 (Anthropic, PBC, 2026): the filing states headcount (about 5,300 at September 30, 2026) and TTM revenue per employee (over $6.6 million); the TTM revenue figure is derived by multiplying the two and is not stated in the filing (`data/anthropic_s1_extract.md`). CoreWeave 10-K (CoreWeave, Inc., 2026a). Cursor: SpaceX acquisition value (Space Exploration Technologies Corp., 2026b), ARR in late April 2026 (Metz, 2026), headcount May 2026 (Business Insider, 2026). Lovable: ARR and headcount February 2026 (Heim, 2026), valuation August 2026 (Bort, 2026). Vercel: valuation September 2025 (Vercel, 2025), run-rate February 2026 (Temkin, 2026), headcount January 2025 (Garfinkle, 2025); the dates do not match, so its ratio is indicative only. Databricks: post-money valuation August 2021 (Databricks, 2021), ARR at end-2021 (Wilhelm, 2022); we found no primary 2021 headcount. Snowflake: valuation at the IPO price (Levy and Picker, 2020), revenue and headcount for the year to January 2021 (Snowflake Inc., 2021). ARR figures are run rates, not reported annual revenue. The CoreWeave valuation is a press estimate.

**Infrastructure layer.** Anthropic's S-1 reports compute and infrastructure costs equal to 48% of revenue in the second quarter of 2026 (Anthropic, PBC, 2026), and CoreWeave carried about $35.1 billion of debt at June 2026, with Microsoft accounting for 67% of its 2025 revenue (CoreWeave, Inc., 2026a,b). Much of the employment this spending supports is at chip makers, hyperscalers, and data-center builders, which are old, large firms. Microsoft offered voluntary exits to about 60,000 employees in October 2026, an eligibility figure rather than an exit count, while guiding calendar-2026 capital expenditure to about $175 billion (TechCrunch, 2026; Microsoft Corporation, 2026); these aggregates are consistent with trading headcount for capital but cannot establish it. Anthropic's own headcount rose from about 940 (December 2024) to about 5,300 (September 2026) as it reported that most of its code was written by its models (Anthropic, PBC, 2026); a single fast-growing firm cannot identify $\sigma_{\text{ai}}$.

**Exits.** Several fast-growing AI firms have been absorbed by large incumbents before reaching scale as independent employers: SpaceX completed its acquisition of Cursor in August 2026 (Space Exploration Technologies Corp., 2026a,b), Inflection's co-founders and team joined Microsoft in 2024 (Microsoft, 2024), and Google licensed Character.ai's technology and hired its co-founders

(Mehta, 2024). In BDS terms, the acquired employment becomes employment at old, large firms, which bears on how the offset ratio would be measured by firm size and age.

# B A Transatlantic Comparison

Prediction 6 predicts a smaller entry-driven employment response in Europe. This appendix checks its premises against data and reports a stylized European parameterization of the model. Table 9 summarizes the comparison; computations and sources are in `data/europe_evidence.md`.

Table 9: United States vs. Europe: entry, finance, regulation, and labor-market rigidity

| Dimension | United States | European Union |
|---|---|---|
| *Business formation* | | |
| Employer start-up rate | 9.1% (2023) | 4.8% (2024); 5.2% (2021) |
| in ICT / Information | 12.0% (NAICS 51, 2022) | 6.4% (NACE J, 2024); 7.7% (2022) |
| Births with no employees | n/a (nonemployer series) | 84% of all births (2024) |
| *AI finance and adoption* | | |
| Private AI investment, 2025 | \$285.9B | \$20.9B (ratio $\approx$ 14); 9–12$\times$ in Crunchbase/Dealroom |
| AI use, large vs. small firms | 1.7$\times$ (250+ vs. 1–4) | 3.2$\times$ (250+ vs. 10–49, 2025) |
| *Entry costs and regulation* | | |
| Admin. burden on start-ups (PMR, 0–6) | 2.03 | 1.11 (SE) – 2.74 (DE); FR 1.60, IT 1.80 |
| AI-specific horizontal regulation | None at federal level | AI Act; high-risk duties from Dec 2027 |
| *Labor-market rigidity* | | |
| EPL, regular contracts (0–6) | 0.77 | 1.94 (DK) – 2.88 (NL); DE 2.33, FR 2.68, IT 2.86 |

*Sources:* US start-up rates from the BDS, 1978–2023 release (U.S. Census Bureau, 2021); EU employer start-up rate = births with $\geq 1$ employee $\div$ active enterprises with $\geq 1$ employee, business economy excluding holding companies (Eurostat, 2026a). AI investment from Stanford Institute for Human-Centered Artificial Intelligence (2026), with Dealroom.co (2026) as a cross-check; adoption ratios from BTOS (U.S. Census Bureau, 2026b) and Eurostat (2026b) (the EU survey excludes firms with fewer than 10 persons employed). PMR indicator "administrative requirements for limited liability companies and personally owned enterprises" (OECD, 2024); EPL Version 4, individual and collective dismissals, 2019 (OECD, 2020). Levels are not strictly comparable across the Atlantic (enterprise vs. firm units; persons employed vs. employees).

**Formation, finance, and entry costs.** Europe forms employers at roughly half the US rate: in 2024, 4.8% of EU-27 enterprises with at least one employee were newly born, against a BDS start-up rate of 9.1% in 2023 (Eurostat, 2026a; U.S. Census Bureau, 2021); 84% of European births have no employees. In the EU, employer births in information and communication fell from 7.7% (2022) to 6.4% (2024). Harmonized DynEmp evidence shows that young firms are the main source of net job creation in every country covered, but that the post-entry growth of surviving start-ups varies widely and is tied to national policy (Criscuolo et al., 2014; Calvino et al., 2016); business dynamism has declined across OECD economies, mainly within sectors (Calvino et al., 2020). Private AI investment in 2025 was \$285.9 billion in the United States against \$20.9 billion in Europe (Stanford Institute for Human-Centered Artificial Intelligence, 2026), with the gap concentrated in growth and late-stage rounds (Dealroom.co, 2026). Formal entry costs are not the difference:

on the OECD's administrative-requirements indicator the United States scores 2.03, above France, Italy, Spain, and Sweden (OECD, 2024). Employment protection is: the United States scores 0.77 on the OECD's dismissal index against an OECD average of 2.20 (OECD, 2020), and stricter protection is associated with fewer fast-growing young firms (Bravo-Biosca et al., 2016). The AI Act (European Parliament and Council of the European Union, 2024) adds fixed compliance costs for high-risk uses, with SME provisions that the Digital Omnibus extended to small mid-caps and a deferral of stand-alone high-risk obligations to December 2027 (European Parliament and Council of the European Union, 2026); among EU enterprises that considered AI, 52.5% cited unclear legal consequences as a barrier (Eurostat, 2026b).

**A stylized European parameterization.** We rerun the original (unnormalized) Monte Carlo under three European channels, switched on in turn, holding the $\sigma_{\text{ai}}$ draws, the uniform quantiles of every other prior, and the simulation seeds fixed across scenarios: (i) a potential-entrant pool scaled by 0.53, the ratio of the EU employer birth rate to the US start-up rate; (ii) AI Act compliance, which caps the AI-induced entry-cost reduction at 20% (against 30%) and raises the adoption fixed cost from $\mathcal{U}(2, 10)$ to $\mathcal{U}(3, 12)$; (iii) thinner growth finance, which caps the AI-native entrant productivity shift at 0.10 (against 0.15). Table 10 reports shares of draws under these assumed priors. Among draws in which large-firm labor demand falls, the share with a positive net effect is 61.0% in the US scenario and 43.6% with all European channels on.[1] Most of the gap comes from thinner growth finance (a mean difference in the net effect of −4.3 percentage points), not from the entry pool (−0.2) or AI Act costs (−1.3). This mirrors the main finding that entrant productivity matters far more than entry costs. The model has no firing costs, so the employment-protection channel is omitted, and the mappings from data to parameters are stylized; the comparison has not been rerun under the normalized CES.

Table 10: Shares of draws under US and stylized European parameterizations (original specification)

| Scenario | Share net $> 0$ (all draws) | Share net $> 0$ (large-firm decline) | Median net effect | Mean diff. vs. US (pp, s.e.) |
|---|---|---|---|---|
| United States | 81.7% | 61.0% | +9.7% | — |
| EU: smaller entry pool | 80.5% | 58.0% | +9.8% | −0.2 (0.3) |
| EU: AI Act costs | 79.3% | 57.0% | +8.4% | −1.3 (0.1) |
| EU: thinner growth finance | 73.1% | 50.5% | +5.2% | −4.3 (0.2) |
| EU: all channels | 69.7% | 43.6% | +4.8% | −5.5 (0.3) |

*Notes:* 1,000 draws per scenario with common random numbers; shares of draws under assumed priors, not probabilities. Mean differences are paired differences in the net labor-demand effect, with standard errors. Source: `scripts/europe_comparison.py`, `data/europe_comparison.md`.

[1] The US scenario here uses 1,000 draws, against 2,000 in the main Monte Carlo of Section 5.4, so its shares differ slightly from those reported there.